

NORTHWAY, AK, USA

WMO: 702910

Lat: 62.961N

Lon: 141.929W

Elev: 522

StdP: 95.21

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 26412

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -45.0	(c) -42.0	(d) -37.0	(e) 0.1	(f) -33.4	(g) -36.1	(h) 0.1	(i) -32.9	(j) 6.9	(k) -19.5	(l) 5.5	(m) -17.8	(n) 0.1	(o) 80	(p) 0.712

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.2	25.7	14.8	23.8	14.0	22.0	13.3	15.9	23.6	15.0	21.9	14.1	20.1	2.3	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.0	10.0	16.5	12.2	9.4	16.0	11.3	8.9	15.5	46.2	23.5	43.5	22.0	40.9	20.3	25.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6.4	5.4	4.7	-45.8	28.5	3.4	1.8	-48.2	29.8	-50.2	30.9	-52.1	31.9	-54.5	33.2
			-37.1	18.3	2.1	2.6	-38.7	20.2	-39.9	21.7	-41.1	23.2	-42.7	25.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-4.3	-25.4	-19.9	-13.7	-1.0	8.0	13.6	15.1	12.3	6.1	-5.0	-18.3	-24.0	(6)
(7)		DBStd	15.90	8.87	7.73	7.51	5.56	4.09	3.11	2.49	3.14	3.81	5.88	7.67	7.23	(7)
(8)		HDD10.0	5594	1096	837	734	331	88	4	1	12	122	465	849	1054	(8)
(9)		HDD18.3	8256	1354	1071	992	581	320	146	103	187	366	723	1099	1312	(9)
(10)		CDD10.0	389	0	0	0	0	27	112	159	85	6	0	0	0	(10)
(11)		CDD18.3	8	0	0	0	0	0	4	3	1	0	0	0	0	(11)
(12)		CDH23.3	208	0	0	0	0	11	77	81	40	0	0	0	0	(12)
(13)		CDH26.7	29	0	0	0	0	1	16	9	4	0	0	0	0	(13)
(14)	Wind	WSAvg	1.4	0.5	0.8	1.4	1.8	2.3	2.1	1.9	1.7	1.6	1.3	0.7	0.6	(14)
(15)	Precipitation	PrecAvg	244	7	7	4	5	24	48	61	37	27	13	8	7	(15)
(16)		PrecMax	386	26	31	15	18	66	112	126	75	89	34	27	20	(16)
(17)		PrecMin	144	0	0	0	0	1	10	7	8	0	2	1	2	(17)
(18)		PrecStd	57	5	7	4	5	18	23	31	19	19	8	6	4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-2.6	-1.6	7.4	17.2	24.9	29.2	27.9	27.2	19.2	9.9	-1.8	-5.7	(19)
(20)			MCWB	-3.6	-3.2	2.4	8.2	12.1	16.4	15.7	15.9	10.6	4.8	-2.4	-6.1	(20)
(21)		2%	DB	-8.3	-4.4	4.0	12.5	22.0	25.3	25.6	24.1	17.1	6.4	-4.2	-10.1	(21)
(22)			MCWB	-8.7	-5.5	0.0	5.3	11.5	14.3	15.1	15.0	9.7	3.0	-4.7	-10.7	(22)
(23)		5%	DB	-11.4	-7.1	1.2	10.3	19.4	23.2	23.6	21.6	15.2	3.8	-6.4	-12.2	(23)
(24)			MCWB	-11.8	-7.8	-1.8	4.0	9.8	13.3	14.5	13.7	8.8	1.5	-6.8	-12.6	(24)
(25)		10%	DB	-13.8	-9.3	-1.4	8.5	17.1	21.2	21.9	19.3	13.5	1.9	-8.3	-14.0	(25)
(26)			MCWB	-14.3	-9.9	-3.7	2.9	8.7	12.6	13.8	12.5	8.1	0.5	-8.7	-14.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-3.2	-3.0	2.9	8.6	13.5	17.7	16.7	16.9	11.7	5.6	-2.4	-5.7	(27)
(28)			MCDWB	-2.6	-1.6	7.0	16.2	21.9	25.3	24.7	24.7	17.0	8.4	-1.8	-5.1	(28)
(29)		2%	WB	-8.1	-5.3	0.2	5.8	11.7	15.7	15.8	15.4	10.5	3.2	-4.6	-10.1	(29)
(30)			MCDWB	-7.8	-4.2	3.6	12.0	20.9	23.8	23.3	22.4	15.4	5.9	-4.1	-9.7	(30)
(31)		5%	WB	-11.2	-7.5	-1.7	4.2	10.4	14.2	15.1	14.3	9.6	1.8	-6.8	-12.3	(31)
(32)			MCDWB	-10.8	-6.8	1.3	9.5	18.1	21.2	21.9	20.1	14.1	3.5	-6.5	-11.9	(32)
(33)		10%	WB	-13.5	-9.7	-3.7	3.1	9.3	13.1	14.4	13.1	8.6	0.5	-8.5	-14.0	(33)
(34)			MCDWB	-13.0	-9.1	-1.3	7.9	16.1	19.6	20.5	18.2	12.8	1.7	-8.2	-13.6	(34)
(35)	Mean Daily Temperature Range	MDBR		8.3	11.6	16.6	14.2	12.9	12.1	11.2	11.1	8.4	7.9	8.0	(35)	
(36)		5% DB	MCDBR	9.0	12.0	17.6	15.5	17.1	15.6	15.0	15.4	14.5	10.7	7.0	8.9	(36)
(37)			MCWBR	8.8	11.0	13.7	9.1	7.6	6.4	6.2	7.1	7.9	7.5	6.8	8.7	(37)
(38)		5% WB	MCDBR	8.9	11.7	16.6	14.6	15.6	13.7	12.8	13.6	12.5	9.5	7.0	8.9	(38)
(39)			MCWBR	8.7	10.8	12.8	8.7	7.1	6.4	5.6	6.6	7.2	6.8	6.9	8.8	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.201	0.242	0.259	0.333	0.349	0.384	0.377	0.372	0.299	0.267	0.187	0.170	(40)
(41)		taud		1.997	2.247	2.271	2.172	2.391	2.307	2.357	2.378	2.577	2.477	2.058	1.924	(41)
(42)		Ebn at Noon		449	731	859	843	866	835	831	801	821	707	478	271	(42)
(43)		Edh at Noon		32	57	83	116	103	114	106	95	64	46	28	16	(43)
(44)	All-Sky Solar Radiation	RadAvg		0.23	0.97	2.72	4.56	5.28	5.45	4.91	3.95	2.49	1.10	0.33	0.10	(44)
(45)		RadStd		0.02	0.08	0.18	0.27	0.24	0.38	0.33	0.33	0.20	0.12	0.03	0.01	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.90	+2.24	N/A	N/A	N/A	N/A	-298	-338	N/A	N/A
(47) Regional (1 neighbor)		+0.77	+2.40	+1.29	N/A	N/A	N/A	-265	-282	+51	N/A

Nomenclature: See separate page